

Eenheid 4.3

Note Title

2015/03/26

Hoe die afgeleide die vorm van die grafiek beïnvloed. p 290

Definisie: f stygend / dalend op Click up

• Stygende / Dalende tests - Click up

→ Hoe bepaal jy die interval I ?

b) as $f'(x) < 0$ op in interval I , dan is f dalend op I

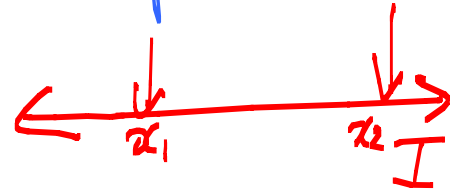
Te bewys: f is dalend op I dws. $f(x_1) > f(x_2)$

Bewijs: do $f'(x_1) < 0$ op I beteken dat f is
differentieelbaar op I

uit ~~stelling~~ beteken dat dat f continu is op I .

Laat $x_1, x_2 \in I$ met $x_1 < x_2$

dan is f continu op $[x_1, x_2]$
en f is diffb. op (x_1, x_2)



Volgens MWS bestaan $c \in (x_1, x_2)$

$$\text{sodat } f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$f(x_2) - f(x_1) = f'(c) (x_2 - x_1)$$

Nou gegee $x_1 < x_2 \Rightarrow x_2 - x_1 > 0$
en $f'(c) < 0$ (gegeve)

$$\text{Dus } f(x_2) - f(x_1) < 0 \Rightarrow -f(x_1) < -f(x_2) \\ \Rightarrow f(x_1) > f(x_2)$$

$\Rightarrow f$ is dalend op I

VBD Bepaal die intervallen waarop $f(x) = xe^x$ dalend is.

$$f(x) = xe^x$$

$$\Rightarrow f'(x) = 1e^x + xe^x = e^x(1+x)$$

f dalend : moet aantonen waar $f'(x) < 0$

$$\Rightarrow \underbrace{e^x}_{>0} \underbrace{(1+x)}_{<0} < 0$$

$$\Rightarrow e^x > 0 \text{ en } 1+x < 0$$

$$\Rightarrow x \in \mathbb{R} \text{ en } x < -1 \Rightarrow x < -1$$

$f'(x) < 0$ op $(-\infty, -1)$
dus f is dalend op $(-\infty, -1)$

no 13 p 297

Bepaal de intervallen waarop f stijgt daal

ao $f(x) = \sin x + \cos x$, $x \in [0, 2\pi]$

- f is continu op $[0, 2\pi]$
- f is diff op $[0, 2\pi]$

• $f'(x) = \cos x - \sin x$

$f'(x) = 0 \Rightarrow \cos x = \sin x$

$\Rightarrow 1 = \frac{\sin x}{\cos x} = \tan x$

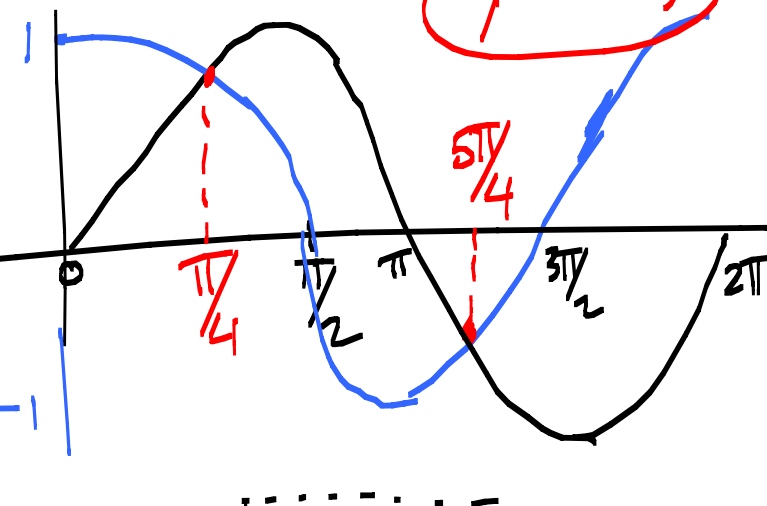
$x = \frac{\pi}{4}$ of $x = \frac{5\pi}{4}$

x -1

x

$y = f(x)$

$y = f(x)$



- f stijgt ao $f'(x) > 0$:

$f'(x) = \cos x - \sin x > 0 \Rightarrow \underline{\cos x} > \underline{\sin x}$

$f'(x) > 0$ op $(0, \frac{\pi}{4})$ en $(\frac{5\pi}{4}, 2\pi)$

Dus f styg op $(0, \frac{\pi}{4})$ en $(\frac{5\pi}{4}, 2\pi)$

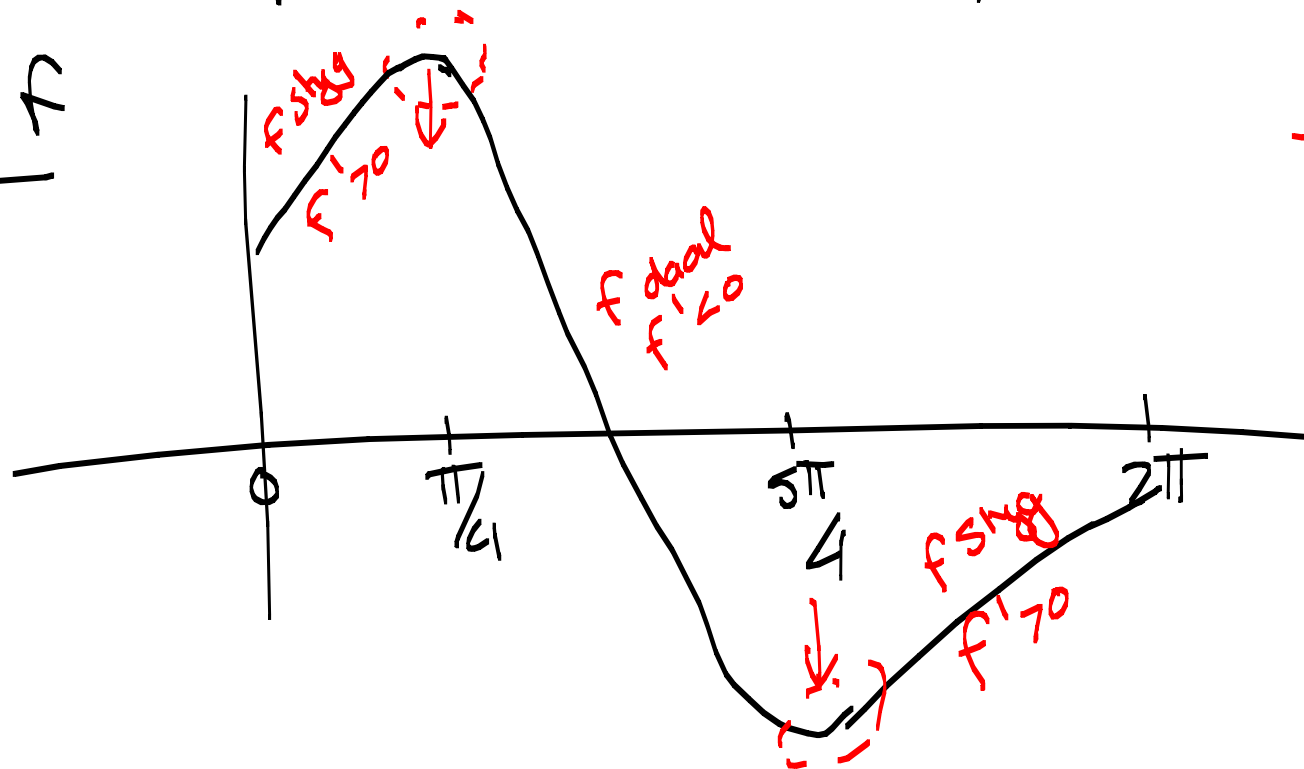
- f daal as $f'(x) < 0$:

$$f'(x) < 0 \Rightarrow \cos x < \sin x$$

$$\Rightarrow f'(x) < 0 \text{ op } (\frac{\pi}{4}, \frac{5\pi}{4})$$

Dus f daal op $(\frac{\pi}{4}, \frac{5\pi}{4})$

Skets f



f' verander van
teken
by lokale
ekstrema

Sien ClickUp om Eerste Afgeleide Toets

⑦ Bepaal de lokale extreme van $f(x) = -x^5 + 15x^3$

- f is continu op $\mathbb{R}$ want f is in polynoom

- kritieke getalle

* $f'(x)$ bestaan oral

$$\begin{aligned} * f'(x) = 0 &\Rightarrow -5x^4 + 45x^2 = -5x^2(x^2 - 9) \\ &= -5x^2(x-3)(x+3) = 0 \end{aligned}$$

$\Rightarrow x=0$ of $x=3$ of $x=-3$
kritieke getalle is $c=0, c=3, c=-3$

- Teken van f'
GEBRUIK 'N GETALLELYN

f' neg pos pos neg $f'(x) = -5x^2(x-3)(x+3)$
-3 0 3

$(-4) \uparrow \uparrow \uparrow$
 $(-)(-)(-)$
 $(-)(-)(+)$
 $(-)(-)(+)$
 $(-)(+)(+)$

kies n getal $x < -3$

$f(-4) = -x - x - = -$

$f(-1) = -x - x + = +$

$f(1) = -x - x + = +$

$f(4) = -x + x + = -$

• Lokale extreme

* $f' < 0$ op $(-\infty, -3)$ $\therefore f$ daal op $(-\infty, -3)$

$f' > 0$ op $(-3, 0)$ $\therefore f$ stijgt op $(-3, 0)$

$x = -3$ $\therefore f$ het in lokale min $f(-3) = -162$ in $x = -3$

WANT f' verander van neg na pos

* By $x = 0$
 f' verander nie van teken nie

$\therefore$ Geen lokale ekstremum by $x = 0$ nie

* By $x = 3$

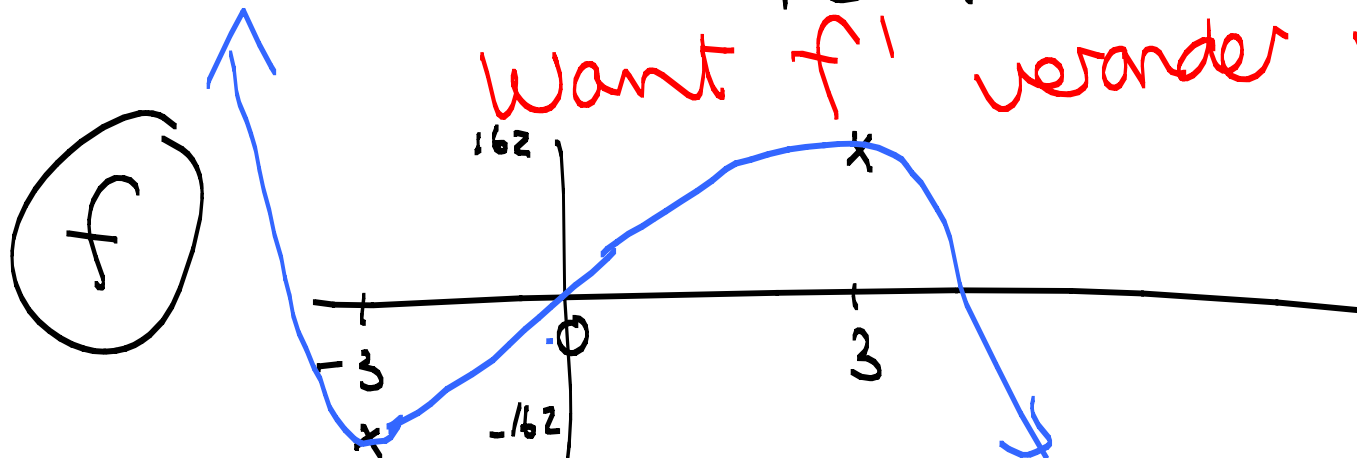
$f'(x) > 0$ op $(0, 3)$ $\therefore f$ styg op $(0, 3)$

$f'(x) < 0$ op $(3, \infty)$ $\therefore f$ daal op $(3, \infty)$

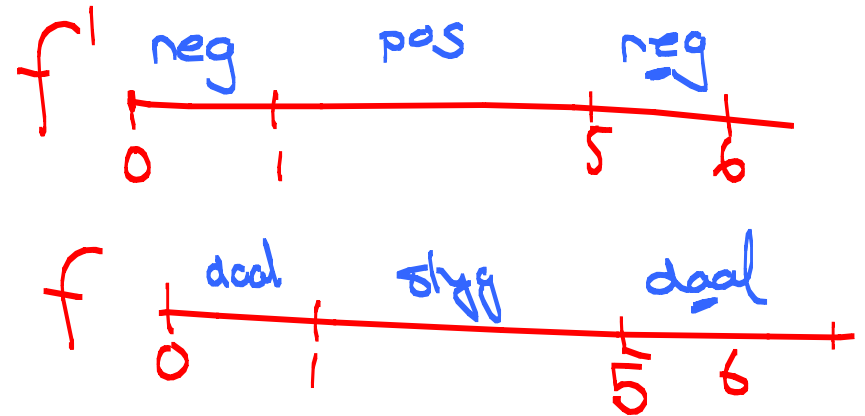
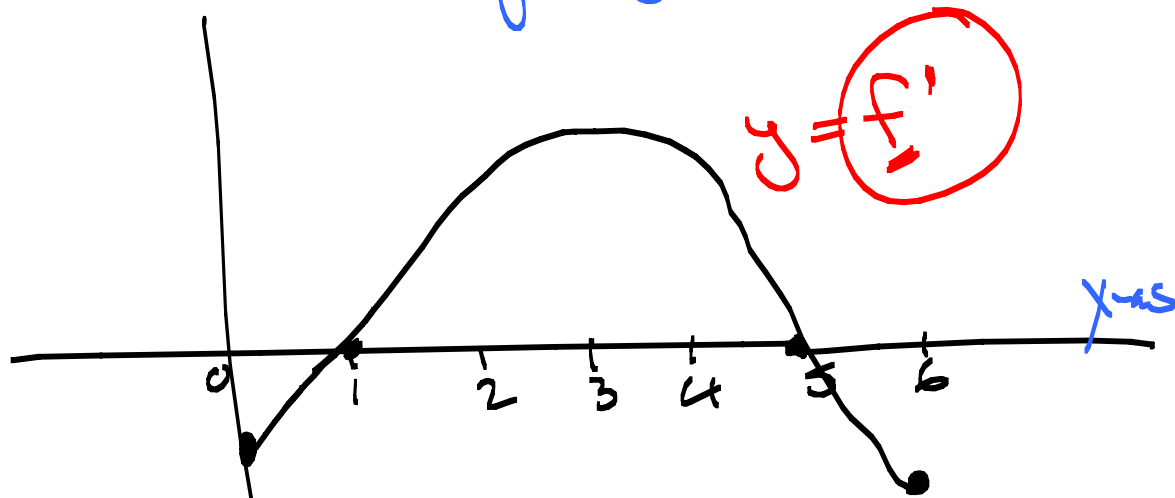
$\therefore f$ het 'n lokaal maks by $x = 3$

$$\text{en } f(3) = 162$$

Want f' verander van pos na neg



② Vind die lokale ekstreme van f as die grafiek van f' gegee is



By $x=1$ het f 'n lokale minimum en $f(1)$

By $x=5$ het f 'n lokale maksimum en $f(5)$

Op ClickUP

Definisies :

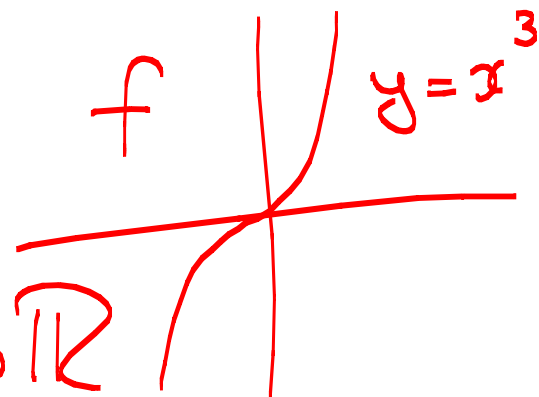
- f konkav na bo
- f konkav na onder

- toets vir konkaaf na bo / ander

VBD $f(x) = x^3$ (Toets vir konkaafheid)

$$f'(x) = 3x^2$$

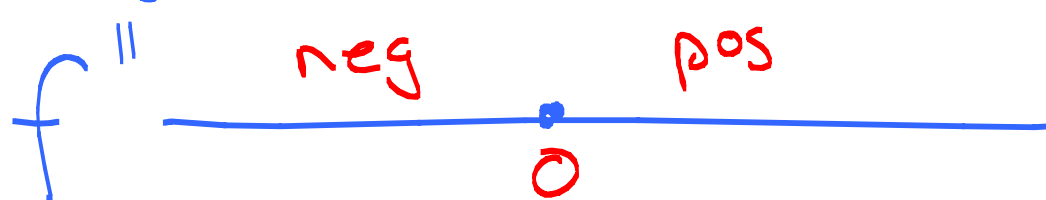
$$f''(x) = 6x$$



- f is 'n stygende funksie op $\mathbb{R}$

Dan is $f'(x) > 0$ vir alle $x \in \mathbb{R}$

Getallelyn: (Tekens van f'')



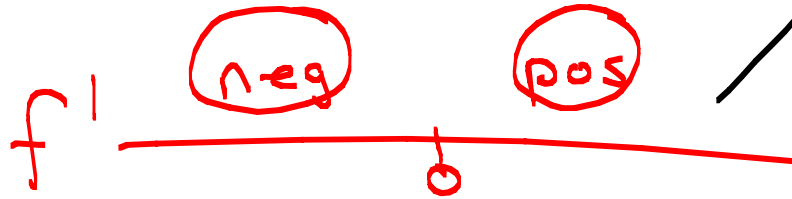
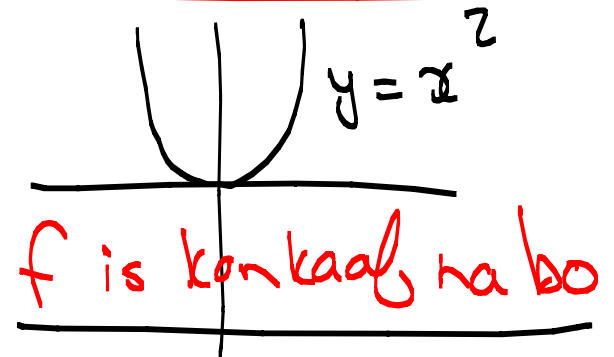
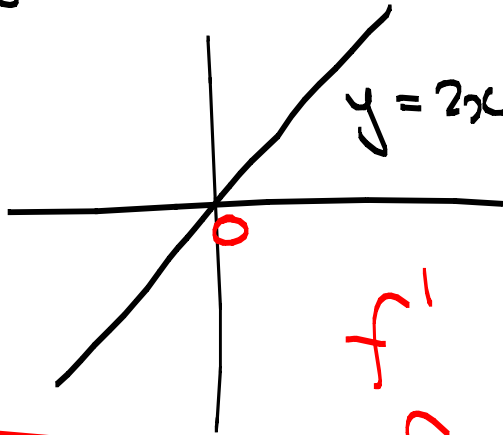
$$f''(x) = 0 \\ \Rightarrow \underline{6x} = 0 \\ x = 0$$

$$\begin{aligned} \bullet f''(-1) &= -6 \\ f''(1) &= 6 \end{aligned}$$

f is konkkaaf na onder op $(-\infty, 0)$
en f is konkkaaf na bo op $(0, \infty)$

Redenasie: $y = f(x) = x^2$

• $y' = f'(x) = 2x$



f' is 'n stygende
funksie

f styg
 $f' > 0$ ✓

f' styg
 $f'' > 0$

Dus $f''(x) > 0 \therefore f$ is konkkaaf na bo

ClickUP : Definitie van Inflectionpunt.

VBD : $f(x) = 1 + 3x + 3x^2 + x^3$

f is continu op $\mathbb{R}$

$$f'(x) = 3 + 6x + 3x^2$$

$$f''(x) = 6 + 6x = 6(1+x)$$

Teken van f'' : $f''(x) = 0 \Rightarrow x = -1$

f''

neg

pos

x	-2	0
$f''(x)$	-	+

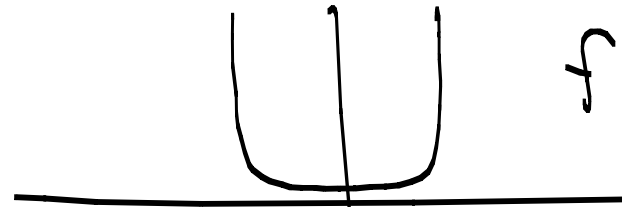
$f''(x) < 0$ op $(-\infty, -1)$ $\therefore f$ is konkav na order op $(-\infty, -1)$

$f''(x) > 0$ op $(-1, \infty)$ $\therefore f$ is koncaaf na bo op $(-1, \infty)$

$\therefore f$ het 1 inflexiepoint by $x = -1$

Click up Determinatie van 2de afgeleide Toets

VBD $f(x) = x^4$



f is koncaal na
0

- $f'(x) = 4x^3$

kritieke getal: $f'(x) = 0 \Rightarrow x = 0$

$f'(\text{red}) = 0$ ✓

- $f''(x) = f''(0) = 12(0)^2 = 0$

$f''(x) = 12x^2$

$\therefore f'' \not> 0$

$f'' \not< 0$

kan nie 2de afgeleide Toets
gebruik nie

Gebruik 2de afgeleide tests en bepaal de lokale extreme van f

1) $f(x) = x e^{3x}$

• $Df = \mathbb{R}$

• $f'(x) = 1x e^{3x} + 3x e^{3x} = e^{3x} (1 + 3x)$

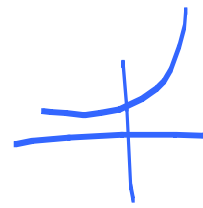
• $f''(x) = 3e^{3x} (1 + 3x) + e^{3x} (3) = 3e^{3x} ([1 + 3x] + 1) = 3e^{3x} (3x + 2)$

• $f'(x) = 0 \Rightarrow e^{3x} (1 + 3x) = 0$

$\Rightarrow x = -\frac{1}{3}$

$\therefore f'(-\frac{1}{3}) = 0$

$f''(-\frac{1}{3}) = 3e^{3(-\frac{1}{3})} (3(-\frac{1}{3}) + 2) = 3e^{-1} (-1 + 2) = 3e^{-1} = \frac{3}{e}$



Now is $f''(-\frac{1}{3}) = \frac{3}{e} > 0$

$\therefore f$ hat lokale min bei $x = -\frac{1}{3}$

n1. $f(-\frac{1}{3}) = (-\frac{1}{3})e^{3(-\frac{1}{3})} = -\frac{1}{3e}$
 $(f(x) = xe^{3x})$

② $f(x) = \frac{\ln x}{x}$

• $D_f = (0, \infty)$

• $f'(x) = \frac{\frac{1}{x} \cdot x - \ln x \cdot 1}{x^2} = \frac{1 - \ln x}{x^2}$

• $f''(x) = \frac{(-\frac{1}{x})x^2 - (1 - \ln x)(2x)}{x^4}$

$= \frac{-x - 2x + 2x \ln x}{x^4} = \frac{x(-1 - 2 + 2 \ln x)}{x^4}$

$$= \frac{-3 + 2 \ln x}{x^3} *$$

$$f'(x) = 0 \Rightarrow \frac{1 - \ln x}{x^2} = 0 \Rightarrow 1 - \ln x = 0$$

$$f'(e) = 0 \checkmark \Rightarrow e^{\ln x = 1} \Rightarrow x = e$$

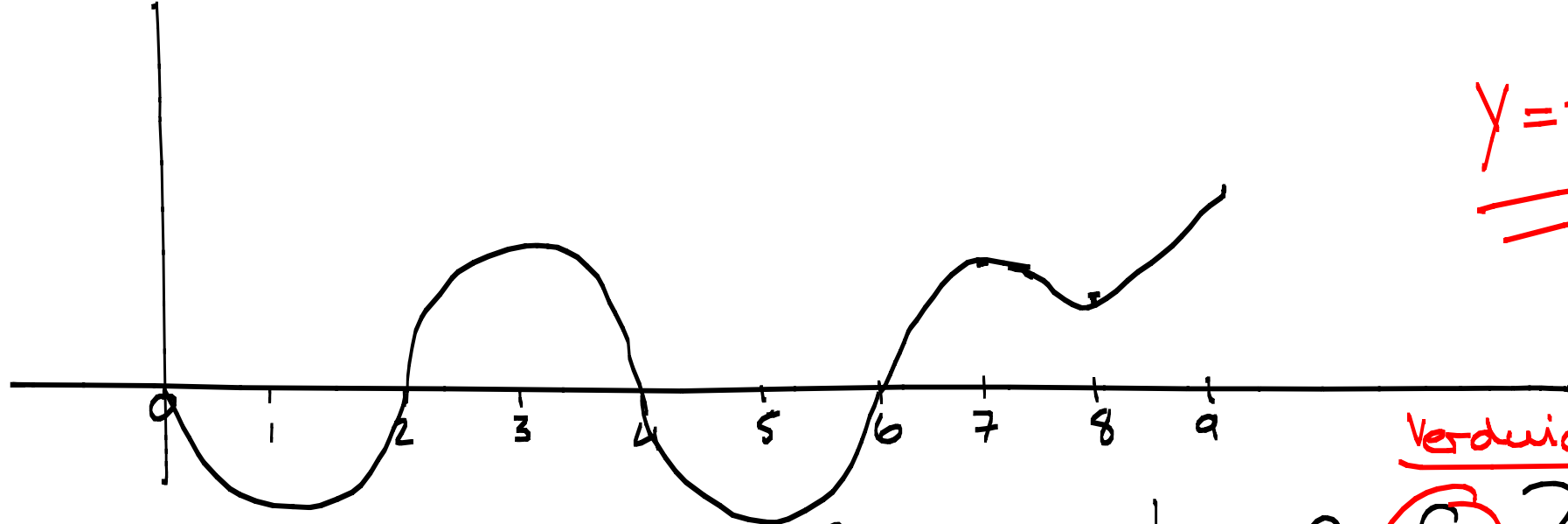
$$f''(e) = \frac{-3 + 2 \ln e}{e^3} = \frac{-1}{e^3} < 0$$

$\therefore f''(e) < 0 \checkmark \therefore f$ has a local max by

$$x = e \text{ n.l. } f(e) = \frac{\ln e}{e} = \frac{1}{e}$$

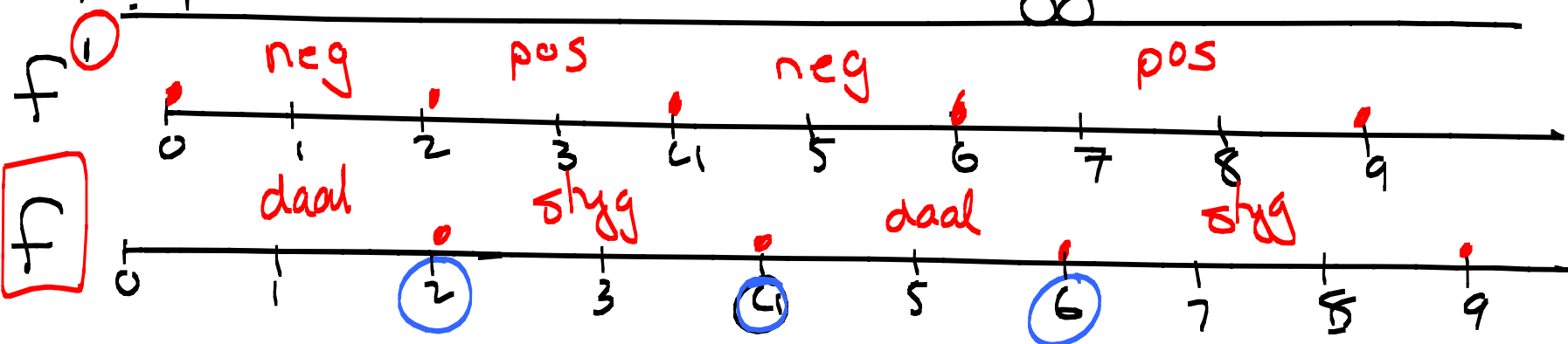
NB VBD Ex 4.3 p 298 no 8

$$\underline{\underline{y = f'(x)}}$$



Verduidelik antw.

a) op welke intervallen styg / daal f ?



f daal op $(0,2)$ en $(4,6)$ Want $f'(x) < 0$ op $(0,2)$ en $(4,6)$

f styg op $(2,4)$ en $(6,9)$ Want $f'(x) > 0$ op $(2,4)$ en $(6,9)$

b) Lokale extreme van f : ✓

f het in lokale minimum $f(2)$ by $x=2$

WANT f' verander van neg na pos by $x=2$

f het in lokale maksimum $f(4)$ by $x=4$

want f' verander van pos na neg by $x=4$

f het in lokale minimum $f(6)$ by $x=6$

want f' verander van neg na pos by $x=6$
